EnvironmSafe for Engineering
Trading, and services

Inspection & Testing Services

INSPECTION REPORT

External visual inspection – V-201 Air receiver / pressure vessel

Client

DEMO – Sample Process Plant (not a real client)

Demo Utilities Area

Document control	
Document number	ES/DEMO-EXT-2026-001
Revision	Rev. 00
Date	10 Oct 2026
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Inspection date	10 Oct 2026
Prepared by	Demo Inspector
Reviewed by	Demo Office Lead
Approved by	General Manager

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1. DOCUMENT CONTROL

Rev.	Date	Description	Prepared	Reviewed	Approved
00	11 Oct 2026	First issue	Demo Inspector	Demo Office Lead	General Manager

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2. SCOPE

The scope of work covered the inspection of the equipment listed in this report in accordance with the referenced codes and the client's requirements.

This report records the external visual inspection of V-201 (Air receiver / pressure vessel) at Demo Utilities Area, which we carried out on 10 Oct 2026 with reference to API 510 (2022 edition). The inspection was witnessed by Demo Plant Engineer.

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3. EQUIPMENT DETAILS

Tag number	V-201	**Description**	Air receiver / pressure vessel
Equipment type	Pressure vessel	**Site / location**	Demo Utilities Area – Compressor house
Manufacturer	Demo Fabricators	**Model**	AR-10
Serial number	DF-5521	**Year built**	2012
Service	Compressed air	**Code of construction**	ASME VIII-1
Material	SA-516 Gr 70	**Capacity**	10 m³
Design pressure	12 bar	**MAWP**	12 bar
Design temperature	80 °C	**Orientation**	Vertical
Corrosion allowance (mm)	3		

4. METHOD

We carried out a close visual examination of the accessible external surfaces, welds, nozzles, supports and attachments, using hand lamps, mirrors and gauges where needed. Conditions found were recorded and photographed.

Checklist

No.	Item	Result	Note
1	Nameplate present and legible	Yes	
2	Shell / body free of visible corrosion, dents or bulges	No	Corrosion at bottom head
3	Supports, saddles, foundations and anchor bolts in good condition	No	Pitting at north leg
4	No leaks observed	Yes	

5. RESULTS

We carried out an external visual inspection and an ultrasonic thickness survey of the air receiver. External corrosion at the bottom head has progressed since the previous inspection. The drain valve has been replaced. The origin of the pitting at the north support leg could not be established from the as-found condition.

6. THICKNESS READINGS

Point	Location	Nominal (mm)	t-min (mm)	Previous (mm)	Measured (mm)	ST rate (mm/y)	LT rate (mm/y)	Rem. life (y)
CML-01	Top head	12	8.2	11.6 (2025)	11.5	0.100	0.060	33.0
CML-04	Bottom head	12	8.2	10.6 (2025)	10.1	0.500	0.219	3.8
CML-05	Drain nozzle	7.1	4	6 (2025)	5.8	0.200	0.139	9.0

ST = short-term rate (last two readings); LT = long-term rate (first and last readings). Remaining life uses the higher of the two rates.

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7. STATUS OF PREVIOUS FINDINGS

Ref.	Finding	Raised	Previous status	Status now	Our note
F-001	External corrosion at bottom head	10 Oct 2025	Open	Open – deteriorated	Pitting deeper; coating breakdown now approx. 45 % of the head.
F-002	Drain valve passing	10 Oct 2025	Open	Closed	Valve replaced by client in March 2026.
F-003	Nameplate partly illegible	10 Oct 2025	Open	Open – unchanged	No change.

8. FINDINGS AND PRIORITIES

Ref.	Finding and location	Severity	Priority	Recommendation	Due
F-004	Pitting at north support leg – North leg, 300 mm above base plate Scattered pitting up to an estimated 1.5 mm deep over an area of approx. 150 × 200 mm.	High	P1	Clean to bare metal, measure pit depth and assess against API 579-1 Part 6; restore coating.	10 Apr 2027
F-005	Earthing strap loose	Low	P3	Re-tighten and test the earthing connection.	—

P1: Action within 6 months. P2: Action at the next shutdown and not later than the set due date. P3: Continuous or routine attention.

9. RECOMMENDATIONS

Restore the coating system on the bottom head and support legs before the next inspection.

- F-004 (P1): Clean to bare metal, measure pit depth and assess against API 579-1 Part 6; restore coating.
- F-005 (P3): Re-tighten and test the earthing connection.

10. CONCLUSION

Fit for continued service with conditions

The equipment is considered fit for continued service subject to the conditions and recommendations stated in this report being completed by the dates given.

Next inspection due	10 Oct 2027
Interval	12 months

11. LIMITATIONS

The internal surfaces were not examined within the scope of this inspection.

This report records the condition of the equipment as found on the date of inspection, within the scope and access available. Areas that were insulated, buried, inaccessible or not prepared were not examined unless stated. Where the origin of a condition cannot be established from the as-found condition, this is stated.

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12. CERTIFICATES

Certificate	Type	Issued	Issued by
ES-H-2026-001	HYDRO	10 Oct 2026	Demo Office Lead

13. SIGN-OFF

Inspected by	Reviewed and issued by
 **Demo Inspector** Senior Inspector · Cert. API 510 #00000 Date: 10 Oct 2026	**Demo Office Lead** Date: 11 Oct 2026

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14. ANNEX – PHOTOGRAPHS

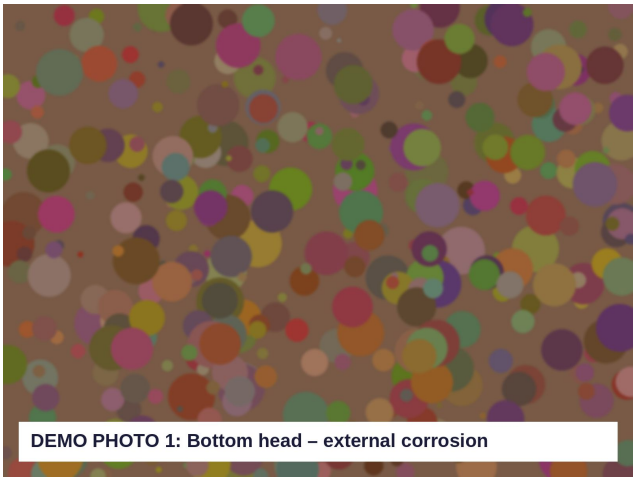


Figure 1 – Bottom head – external corrosion (F-001)

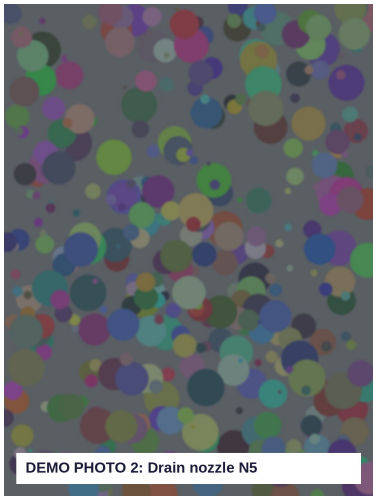


Figure 2 – Drain nozzle N5 with new valve (F-002)

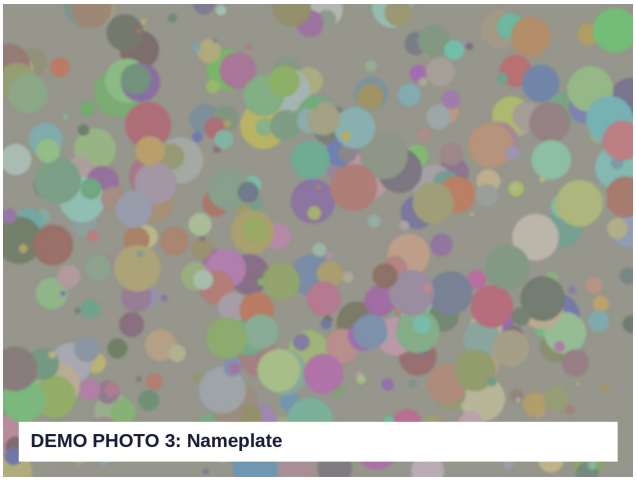


Figure 3 – Nameplate (F-003)

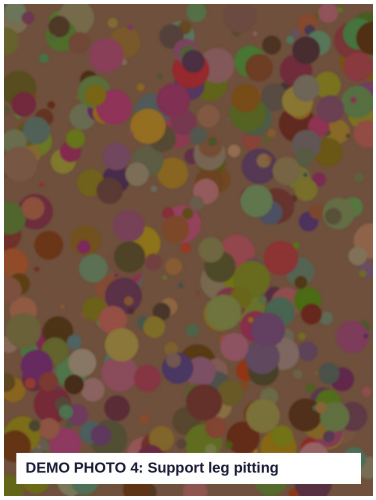


Figure 4 – Pitting at north support leg (F-004)